SAMPLE

Earth's Sky: Atmospheric Optics

Halos, Sundogs, Mirages, and Rainbows

Aira Astralis

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Introduction

The sky is the largest optical instrument ever built, and it charges nothing for the show. On any given day it produces blue noon, gold evenings, rainbows, rings around the sun and moon, phantom lakes on hot roads, pillars of light over cities, and, for a patient observer at the right moment, a brief green flash at the horizon. This book is about that show: the discipline called atmospheric optics, which explains what light does on its way through air, water, and ice to the waiting eye.

Atmospheric optics is the study of light in the wild. Its laboratory is the atmosphere itself, its apparatus is the full machinery of physical optics, refraction, reflection, dispersion, diffraction, interference, scattering, and polarization, and its subject is every color and shape the sky can make. A convenient distinction, drawn loosely in the older literature, separates atmospheric optics in the broad sense from meteorological optics, the part concerned with what the naked eye can watch without instruments; this book keeps to that older, human scale, while borrowing freely from the modern physics underneath.

The book's claim on the reader's attention is simple: the phenomena are free, universal, and wrongly taken for granted. A person can live an entire life under rainbows, halos, and crepuscular rays without once asking why the secondary bow is dimmer than the primary, or why sun dogs sit at exactly the height of the sun, or why the green flash exists at all. The questions have answers, most of them worked out centuries ago and refined to extraordinary precision since, and the answers belong to the most satisfying kind of science: exact enough to predict a rainbow to a tenth of a degree, and yet visible to anyone who steps outside.

A word on the evidence behind the account. The physics reported here rests on the standard, well-established treatment of the subject, the geometry of the rainbow worked out by Descartes, the wave optics completed by Young and Airy, the scattering theory of Rayleigh and later workers, and the ice-crystal catalog built by a century of patient observers. Where a fact is famous it is stated plainly; where it is a record or a superlative, the phrasing is hedged to what the record actually supports. There are no invented numbers in what follows, and no phenomenon is described that has not been reported by multiple observers.

The road map follows the pattern of the series, so the reader can navigate by chapter number alone. The first chapter tells the history of the subject, from the Greeks through the seventeenth century's geometry to the wave and scattering revolutions of the nineteenth. The second and third chapters are the fundamentals and the catalog: the physics every phenomenon shares, then the phenomena themselves. The fourth chapter goes deeper into the science, the fifth into observation and measurement, the sixth through the signature features, the seventh into the wider context of meteorology, art, and navigation. The eighth reports the modern synthesis, the ninth the active research frontier, the tenth the milestone discoveries, and the eleventh the puzzles that remain. The twelfth looks to the future of the field, the thirteenth collects trivia, and the fourteenth closes with the significance of knowing what the sky is doing. The reader who finishes the book should be able to step outside and see more in an ordinary afternoon than most people see in a year, and should know why they are seeing it.

A word on prerequisites, since the reader deserves to know what a book will ask of them. This one asks almost nothing: no mathematics beyond the idea of an angle, no physics beyond the six actions defined in the second chapter, and no equipment beyond the willingness to look up at the right moments. Where a number appears, it is because the sky's phenomena are honest about their geometry, and the numbers are the fastest way to say what a display looks like; a reader who can hold a protractor in the mind's eye has all the apparatus the subject demands. The chapters can be read in any order after the second, since each is self-contained, and the catalog chapter, the third, is the one most readers will return to; it was written to be consulted in the field, before the fact, and again afterward, when the sky has just shown something and the name of the thing is wanted at once. What the book hopes to install, beyond the explanations, is the observer's reflex, the half-second glance upward at the first sign of ice cloud, the habit of turning around after sunset to face the east, and the instinct to check the space beside the sun on any cold bright day, because most of the subject's displays live within a hand's width of the disk, and most people walk through them for a lifetime without once looking there.

Chapter 1: The Discovery of Atmospheric Optics

The oldest written questions about the sky are questions about its light. Aristotle's Meteorologica, twenty-three centuries ago, collects rainbows, halos, and mock suns in one treatise with the weather, and his successors in the Greek and Roman world kept the habit; Seneca devoted pages of his natural questions to halos and the colors of

the heavens. These early accounts are honest observation wrapped in wrong theory, the rainbow imagined as reflected light from a cloud that behaves like a mirror, and they held for so long precisely because the sky is hard to interrogate: the phenomena appear when they choose, last minutes, and cannot be brought indoors.

The first decisive break came from the Islamic natural philosophers, whose golden-age work on optics generally reoriented the whole discipline. Ibn al-Haytham, working in Egypt in the eleventh century, treated light as a thing with behavior, rays that travel, reflect, and refract by rules, and his great book of optics, translated into Latin in the twelfth century, became the foundation on which European optics was built for four hundred years. It matters for this book's subject because atmospheric light is refraction, reflection, and scattering all at once; a science of the sky could not exist until a science of light did.

The second break came in the seventeenth century, and it came from the rainbow. Descartes, in 1637, did what the ancients never attempted: he followed a ray of sunlight into a raindrop, through its back wall, and out again, applying the law of refraction at every surface, and by tracing enough rays he found the explanation for the bow's size, its height above the antisolar point at roughly forty-two degrees, and of the bright region within. The rainbow, the oldest mystery in the sky, yielded to applied geometry in a single published analysis, and the lesson stuck: the atmosphere's light is mathematics, not mood.

Newton, a generation later, supplied the missing ingredient, the fact that white light is a mixture and that refraction sorts it into colors. With dispersion in hand, the rainbow's bands were explained twice over, by the geometry that gives the bow its size and the physics that gives it its colors. The same century's work on diffraction by Grimaldi, and Huygens' wave picture of light, planted the seed of the third revolution, though the wave had to wait a century and a half for its harvest.

The harvest came in the early nineteenth century, and it came from the rainbow again. Observers had long noticed faint extra bows just inside the main one, the supernumerary arcs, which no ray geometry could explain, since rays were supposed to produce a single caustic, not a striped one. Thomas Young showed in 1804 that interference, the signature of waves, would place bright and dark bands exactly where the arcs appear, and George Airy, in 1838, completed the work with the mathematical theory of the bow that carries his name. The rainbow was now a full wave-optics problem, and its solution marked the moment the atmosphere's light became a proving ground for the deepest theories of light itself.

The nineteenth century also founded the scattering half of the subject. Arago discovered in 1809 that the daylight sky is partially polarized, a fact no one had suspected and no corpuscular theory could easily digest; Babinet and Brewster mapped the neutral points where the polarization vanishes; and by the 1870s Rayleigh had the explanation, scattering by molecules of air, with the famous dependence that blue light scatters far more strongly than red. The blue of the sky, the red of sunset, and the polarization pattern overhead all fell to one law. Meanwhile polar explorers and alpinists were cataloguing ice-crystal phenomena, glories were being described from mountaintops, and a patient Dutch observer tradition was beginning the work of cataloging twilight, dusk colors, and earth's shadow that would culminate in the twentieth century's great field guides.

The twentieth century brought the ice-crystal catalog to completion, largely through the halo observers, who traced the dozens of arcs and rings to the shapes and falls of hexagonal ice, and through the physics of the glory and corona, which belong to wave optics on droplets. The modern era added photography, then cheap digital cameras, and finally the web, which turned a scattered band of solitary watchers into a global, self-reporting network. The discovery history of atmospheric optics is thus a history of instruments shrinking and observers multiplying: from one philosopher squinting at a rainbow to thousands of cameras watching the whole sky at once, with the same sky rewarding both ends of that history.

Two later episodes complete the discovery history, because the subject kept its amateur character even as its theory matured. The green flash passed from sailors' yarn to settled physics only in the closing decades of the nineteenth century and the opening ones of the twentieth, when careful observers began timing and photographing it, and the debate over its very existence, conducted across two centuries between crews who saw it and landsmen who doubted them, is the subject's best illustration of how a visual phenomenon earns scientific standing: by surviving every attempt to explain it away. The Brocken spectre and its glory enjoyed a parallel career in the tourist literature of the nineteenth century, when the mountaineering boom sent thousands of travelers to the German peak that gave the spectre its name, and their descriptions, theatrical, awed, and mostly accurate, gave the physicists of the next century a substantial file of eyewitness accounts to work from. The pattern held in both cases: the observers, unschooled and numerous, gathered the raw record, and the theory, when it finally arrived, found the record waiting. That division of labor, watchers and explainers, amateurs and mathematicians, runs from

Aristotle's correspondents to the modern camera networks, and it is the subject's oldest institution, and its most durable one.